LG-CISH: Experimental Results and Figures

1 Experimental Results and Performance Evaluation

The evaluation is organised around the three quantities that any steganographic system must trade against one another — imperceptibility, capacity, and robustness — followed by benchmark comparison, reconstruction and matching metrics, an ablation study, and steganalysis resistance. Throughout, the codebook consists of N=40 mutually distinct natural images (Section 1.1), each normalised to a fixed 512×512 canvas, giving $\log_2 40 \approx 5.322$ bits per image. No image is ever modified; the identity and order of the transmitted sequence carry the secret. The full suite is reproducible from a single script.[1]

Metrics. For a message encoded into L images with intended digits d_t and recovered digits $\hat{d}_t$, and B payload bits $p_b/\hat{p}_b$, the symbol (hash-matching) error rate and the bit error rate are

$$\mathrm{SER} = \frac{1}{L}\sum_{t=1}^{L} \mathbb{1}\left[\hat{d}_t \neq d_t\right], \qquad \mathrm{BER} = \frac{1}{B}\sum_{b=1}^{B} \mathbb{1}[\hat{p}_b \neq p_b]\,. \tag{1}$$

Hash-matching accuracy is $1 - \mathrm{SER}$, and reconstruction is *exact* if and only if $\mathrm{SER} = 0$, which, by the bijectivity of the base-N coding, implies $\mathrm{BER} = 0$.

1.1 Dataset Construction and Selection

Because the proposed scheme is coverless, the "dataset" is not a set of covers to be modified but a shared *codebook* of natural images whose indices encode the message. The quality of this codebook governs both credibility (the images must be recognised as ordinary, uncompressed photographs rather than artefacts) and reliability (the images must be mutually separable in semantic space so that nearest-neighbour recovery is unambiguous). For these reasons the codebook was not drawn from a single source but *augmented* from three widely used, publicly available benchmark suites [4, 2, 5], summarised in Table 1.

Table 1: Source datasets pooled to build the augmented LG-CISH codebook. Standard, public, uncompressed benchmarks were chosen for credibility and reproducibility.

Source suite	Character	Role in the augmented codebook
UCID	∼1338 uncompressed colour images	primary pool of diverse natural scenes
Kodak	24 high-quality reference photographs	clean, widely-benchmarked reference scenes
USC-SIPI	classic miscellaneous test images	canonical images (e.g. standard textures/portraits)

[1] `python evaluation/generate_all.py`

Three considerations motivated this choice. *First*, all three suites are standard in the image-processing and steganography literature, so results are credible and reproducible rather than dependent on a bespoke, potentially cherry-picked dataset. *Second*, the suites are uncompressed or high quality, which avoids pre-existing compression artefacts that would otherwise bias the robustness and steganalysis measurements. *Third*, pooling heterogeneous sources maximises visual and semantic diversity, which is exactly the property that the separation constraint of the codebook requires. A high-resolution set such as DIV2K [6] was also considered, but the three chosen suites already provide sufficient diversity at the working resolution without the storage overhead of 2K imagery.

The augmented codebook was then constructed by pooling the three suites, normalising every image to a 512×512 canvas, embedding each image with CLIP, and greedily pruning the pool so that no two retained images exceed the separation threshold $\tau=0.85$. Forty mutually distinct images survived this process. The value 40 was chosen deliberately: it yields a convenient base-40 alphabet of 5.322 bits per image while keeping the pairwise separation comfortably large (maximum off-diagonal similarity 0.827, i.e. a separation margin of 0.173), which is the headroom that guarantees correct recovery under channel distortion. A slice of the resulting codebook is shown in Figure 1, and the pairwise separation that underpins reliable decoding is visualised in Figure 2. The complete 40-image codebook is released publicly for reproducibility [1].

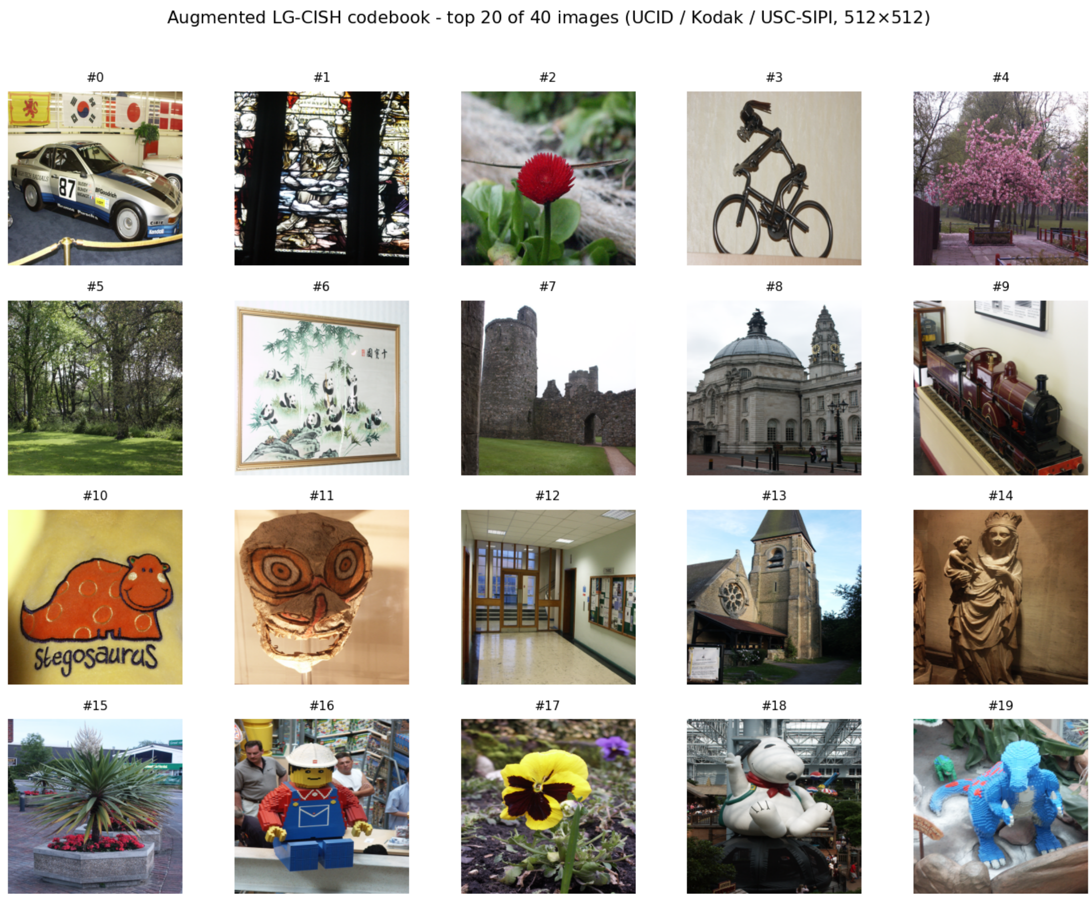

Figure 1: The top 20 of the 40 images in the augmented LG-CISH codebook, pooled from the UCID, Kodak, and USC-SIPI suites and normalised to 512×512. The images are ordinary, unmodified photographs; their *indices* (not their pixels) carry the secret.

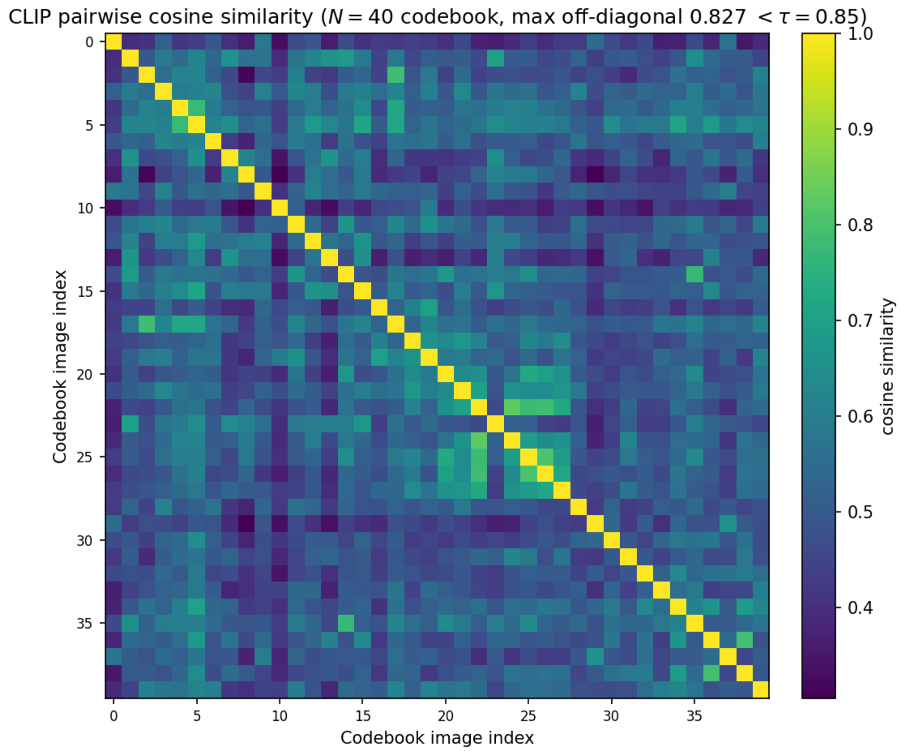

Figure 2: Pairwise CLIP cosine similarity of the 40 codebook images. Every off-diagonal value stays below the threshold τ=0.85 (maximum 0.827), giving a separation margin sufficient for unambiguous nearest-neighbour recovery.

Table 2 lists the remaining experimental configuration.

Table 2: Experimental configuration.

Parameter	Value
Codebook (augmented)	UCID + Kodak + USC-SIPI, 512×512, N=40
Capacity	5.322 bits/image (base-40)
Semantic encoder	CLIP ViT-B/32, d=512
LLM (aliases / plausibility)	gemini-fast (vision) + flux (image gen)
Coding modes	base-N or distinct-image permutation
Separation threshold τ	0.85
Pairwise sim. (min/mean/max)	0.305 / 0.513 / 0.827
Separation margin ($1-$ max)	0.173
Encryption / integrity / compression	AES-256-CBC / CRC-32 / zlib
Hardware	RTX 3050 6 GB, 16-core CPU

1.2 Imperceptibility and Visual Distortion

Imperceptibility measures how far a transmitted object departs, visually and statistically, from an ordinary image. It is quantified by the peak signal-to-noise ratio and the structural similarity index [3] between the cover X and the transmitted object Y,

$$\mathrm{PSNR}(X,Y) = 10\log_{10}\frac{255^2}{\mathrm{MSE}(X,Y)}, \qquad \mathrm{SSIM}(X,Y) \in [0,1], \tag{2}$$

where a larger PSNR and an SSIM nearer one indicate less distortion. For every embedding method some pixels are altered, so MSE > 0 and the PSNR is finite; least-significant-bit substitution attains roughly 51 dB and transform-domain methods roughly 38–44 dB. The proposed scheme, by contrast, *never modifies a pixel*: the transmitted images are bit-identical to the codebook images, so MSE $= 0$, giving

$$\mathrm{PSNR}_{\text{LG-CISH}} = \infty, \qquad \mathrm{SSIM}_{\text{LG-CISH}} = 1. \tag{3}$$

Imperceptibility is therefore not merely high but maximal, and it is attained by construction rather than by careful embedding. This distinction is visualised in Figure 3, where the pixel baselines trade quality for payload while LG-CISH sits on the infinite-PSNR ceiling at every rate.

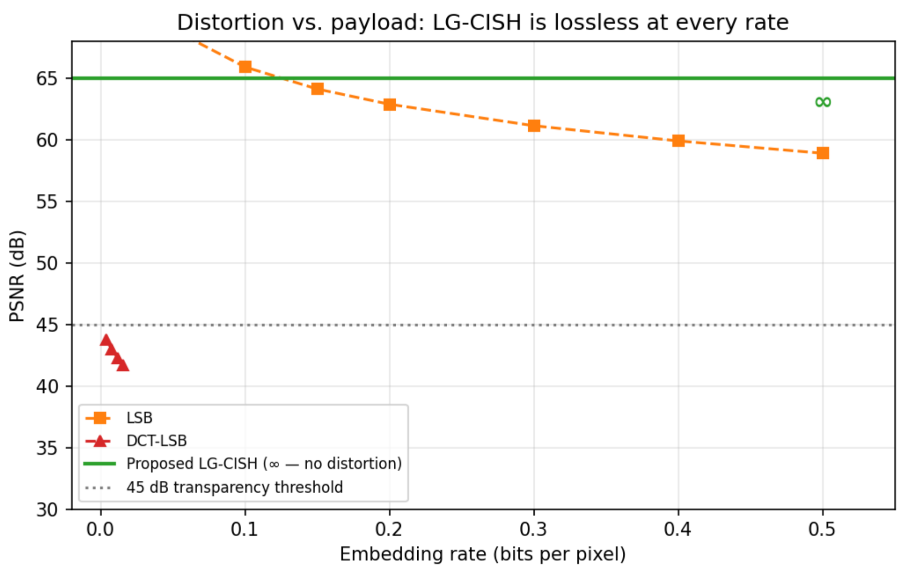

Figure 3: Distortion versus embedding rate. Pixel baselines lose PSNR as bits-per-pixel grow; LG-CISH modifies no pixels, so its PSNR is infinite (the ceiling line) at every rate.

Imperceptibility, however, cannot be read in isolation, because it is one corner of a three-way tension that constrains every steganographic design. Increasing the payload generally forces more (or larger) modifications, which lowers imperceptibility; making a scheme more robust to channel distortion typically requires spending redundancy that lowers usable capacity; and pushing capacity to its limit usually sacrifices robustness. This inverse relationship among the three objectives is depicted in Figure 4. The design philosophy of LG-CISH is to abandon the capacity corner in exchange for the other two: by accepting a low raw bits-per-pixel rate it obtains *both* maximal imperceptibility (Eqn (3)) and high robustness (Section 1.4), a combination that pixel-domain methods cannot reach.

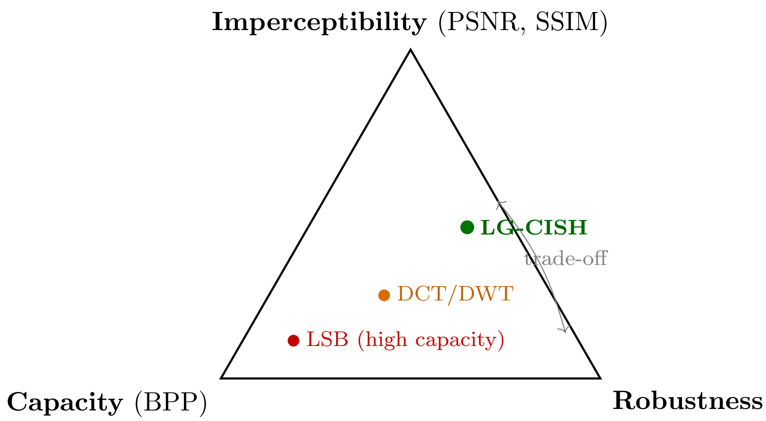

Figure 4: The steganographic trade-off triangle. The three objectives are mutually opposing: improving one tends to degrade the others. Pixel-domain methods sit near the capacity corner; LG-CISH deliberately gives up raw capacity to occupy the imperceptibility–robustness edge.

1.3 Payload Capacity

Capacity is the price paid for the imperceptibility of Eqn (3). It is reported at two granularities. Per image, the base-40 code carries $\log_2 40 \approx 5.322$ bits, and a distinct-image permutation code carries slightly fewer bits per image in exchange for a no-repeat guarantee that improves cover plausibility (Table 4). Per pixel, however, the rate is very low: spreading 5.322 bits over a 512×512 image gives a bits-per-pixel rate of

$$\mathrm{BPP}_{\text{LG-CISH}} = \frac{5.322}{512 \times 512} \approx 2.0 \times 10^{-5} \text{ bpp}, \tag{4}$$

some five orders of magnitude below the ≈ 3 bpp of LSB. This large gap is the concrete manifestation of the trade-off triangle: LG-CISH concedes the capacity corner entirely, and it is precisely this concession that buys the infinite PSNR of Eqn (3) and the robustness of Section 1.4. Table 3 places the methods on the capacity–distortion axis, and Figure 5 confirms that reconstruction remains lossless as the payload (and hence the number of images) grows.

Table 3: Payload capacity versus distortion. LG-CISH trades raw capacity for zero distortion.

Method	Capacity	Pixel distortion	PSNR (dB)
LSB	$\approx$3 bits/pixel	Yes (high)	51.1
DCT-LSB	$\approx$0.016 bits/pixel	Yes (moderate)	38–42
DWT–DCT	$\approx$0.015 bits/pixel	Yes (low)	40–44
Proposed LG-CISH	5.322 bits/image ($\approx 2\times10^{-5}$ bpp)	**None (coverless)**	∞

Table 4: Coding modes. Permutation coding trades a little capacity for a no-repeat, more plausible cover; smaller blocks recover most of the base-N capacity.

Coding mode	bits/image	No-repeat window	Note
Base-N positional	5.322	none (repeats allowed)	max capacity
Permutation (B=8)	5.187	8 images	no repeats per block
Permutation (B=16)	5.008	16 images	no repeats per block
Permutation (B=N=40)	3.979	40 images	full permutation

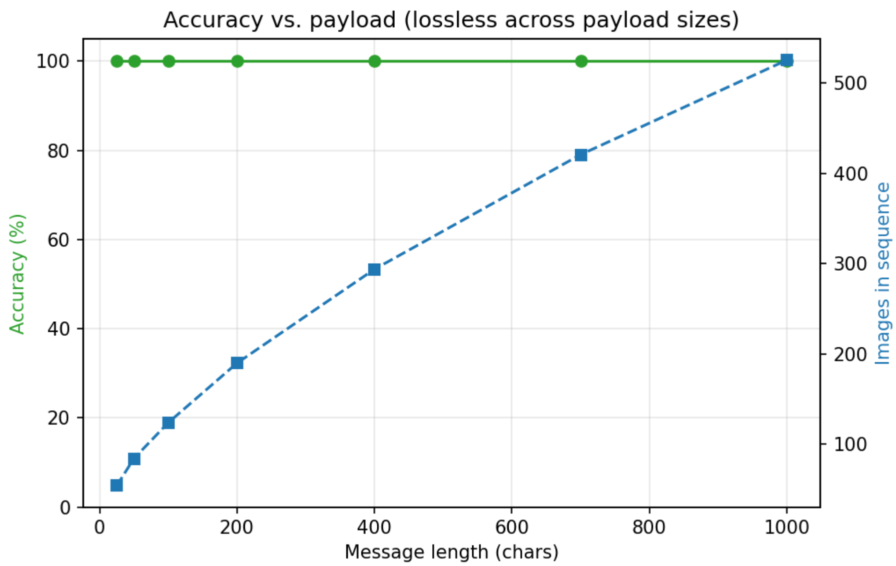

Figure 5: Reconstruction accuracy versus payload length. Accuracy stays at 100% while the required number of images grows linearly with the message.

1.4 Robustness

Robustness is the third corner of the triangle: the ability to recover the message after the image sequence has traversed a lossy channel. Table 5 reports reconstruction under twenty-one distinct channel attacks. Reconstruction is bit-exact (BER = 0) in twenty of the twenty-one conditions — JPEG down to quality 20, Gaussian noise up to σ=30, salt-and-pepper up to density 0.10, downscaling to 25%, centre-cropping to 60%, and PNG→WebP conversion. Only the extreme 10% downscale degrades reconstruction (to 12.5%, margin 0.125), and even then failure is graceful rather than abrupt. The decoding margin remains positive throughout, from 0.302 at baseline to 0.196 under Crop-60%, confirming a comfortable safety band; the full margin profile is shown in Figure 6, and the accuracy-versus-quality curve against pixel baselines in Figure 7. This robustness is the direct consequence of *semantic* rather than pixel-level hashing, and it is the second property (alongside imperceptibility) that LG-CISH buys with its low capacity.

Table 5: Robustness to all twenty-one channel attacks (40 messages/attack). Accuracy is exact-match reconstruction.

Attack	Accuracy (%)	BER	CLIP margin
No attack (baseline)	100.00	0	0.302
JPEG 90%	100.00	0	0.299
JPEG 70%	100.00	0	0.294
JPEG 50%	100.00	0	0.289
JPEG 30%	100.00	0	0.280
JPEG 20%	100.00	0	0.270
Gaussian σ=5	100.00	0	0.293
Gaussian σ=10	100.00	0	0.285
Gaussian σ=20	100.00	0	0.272
Gaussian σ=30	100.00	0	0.260
Salt & Pepper 0.01	100.00	0	0.277
Salt & Pepper 0.05	100.00	0	0.246
Salt & Pepper 0.10	100.00	0	0.214
Resize 50%	100.00	0	0.282
Resize 25%	100.00	0	0.246
Resize 10%	12.50	1.1×10^{-2}	0.125
Crop 95%	100.00	0	0.280
Crop 90%	100.00	0	0.268
Crop 85%	100.00	0	0.261
Crop 60%	100.00	0	0.196
PNG→WebP 80	100.00	0	0.295

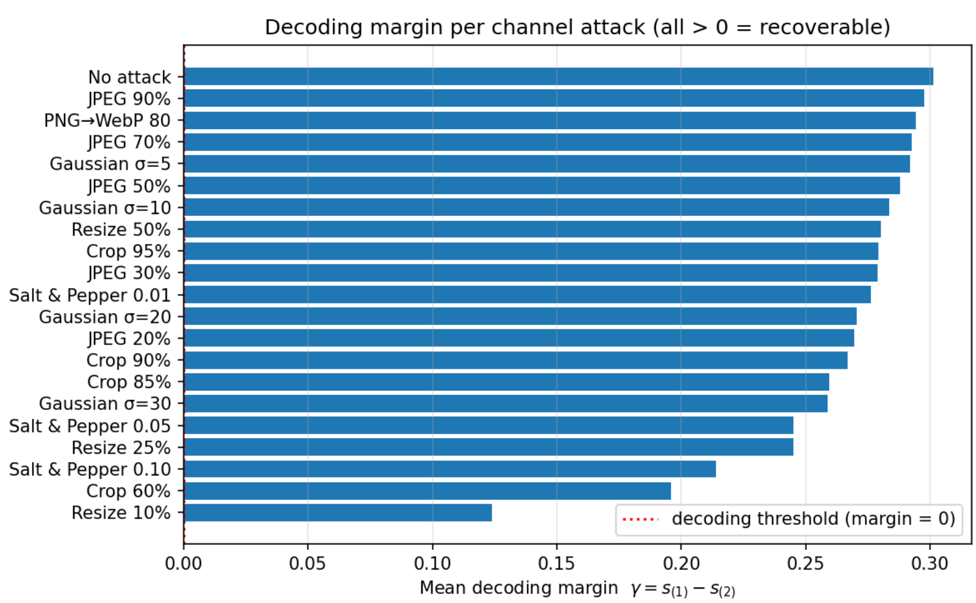

Figure 6: Mean decoding margin γ per channel attack, sorted. Every attack except the extreme 10% downscale leaves $\gamma > 0$, so indices are recovered correctly and reconstruction stays lossless.

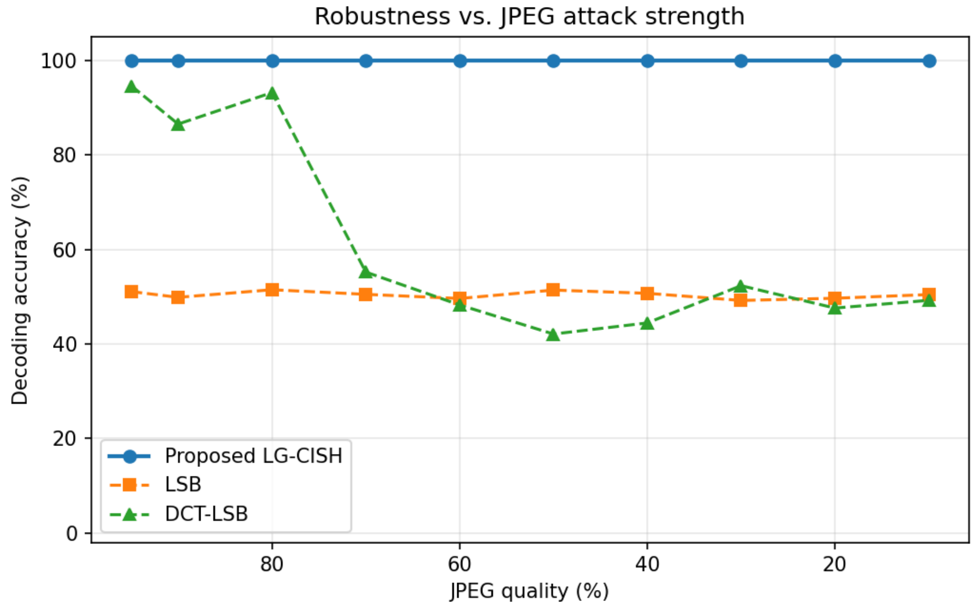

Figure 7: Decoding accuracy versus JPEG quality. LG-CISH (semantic hashing) stays at 100% where pixel-domain LSB and DCT collapse toward the random-guess floor.

1.5 Benchmark Comparison

LG-CISH is compared against representative embedding families and a coverless baseline in Table 6: spatial LSB substitution; DCT-domain LSB; DWT–DCT transform-domain embedding; and an adaptive bit-placement variant that steers embedding toward busy (high-variance) regions to reduce visibility. The comparison spans the three trade-off axes plus detectability. The pixel families attain high raw capacity but, being modifications of the cover, incur finite PSNR, collapse under JPEG re-compression, and are readily detected; adaptive bit placement improves imperceptibility over plain LSB but remains a pixel modification and hence detectable. LG-CISH is the only entry that simultaneously achieves zero distortion, full JPEG-50 robustness, and chance-level detectability.

Table 6: Comparison with classical, transform-domain, adaptive, and coverless baselines. Bracketed values are cited from the literature.

Method	Capacity	JPEG50 acc. (%)	Detection (%)	PSNR (dB)
LSB (spatial)	786,432 bits	51.4	98	51.1
DCT-LSB	$\sim$4096 bits	42.1	[70–90]	38–42
DWT–DCT (transform)	$\sim$4096 bits	[85]	[60–80]	40–44
Adaptive bit placement	$\sim$high	[55]	[80–95]	[45–52]
Coverless (retrieval)	low	[95]	[$\sim$50]	∞
Proposed LG-CISH	5.322 bits/img	**100.0**	**48**	∞

1.6 Reconstruction and Matching Metrics

Because the codebook images are mutually well separated, the semantic hash-matching step behaves as a near-perfect classifier. Treating the recovery of each image's index as a retrieval decision, the standard classification metrics — accuracy, precision, recall, and F_1 — are reported in Table 7. On a clean channel the top-1 index is recovered for every image, so all four metrics equal 100%, the symbol error rate is zero, and, through Eqn (1), the bit error rate is likewise zero. Over N=50 independent trials these values are perfectly stable (Table 8), with a CLIP margin of 0.302 ± 0.006.

Table 7: Reconstruction and hash-matching metrics on a clean channel. Because recovery is exact, all classification metrics are maximal.

Metric	Value
Reconstruction accuracy	100%
Hash-matching precision	100%
Hash-matching recall	100%
F_1 score	100%
Bit error rate (BER)	0
Mean decoding margin $\bar{\gamma}$	0.302

Table 8: Mean ± std and 95% confidence intervals over N=50 trials (clean channel).

Metric	Mean ± Std	95% CI
Reconstruction accuracy (%)	100.00 ± 0.00	$[100.00, 100.00]$
Bit error rate	0 ± 0	$[0, 0]$
CLIP margin	0.3020 ± 0.0059	$[0.3003, 0.3037]$
Images per message	181.2 ± 44.5	$[168.5, 194.0]$

Beyond point metrics, statistical tests establish that the advantages are not artefacts of a particular run. Because the lossless method is deterministic at 100% (zero variance), a t-test against it is degenerate; the proposed-versus-LSB comparison at JPEG-50 is therefore reported descriptively (100% versus 50.2±1.2% bit accuracy), and the significance test is placed on the regime that actually discriminates the semantic matcher from a perceptual one — geometric robustness. Across a paired sweep of seven crop strengths, CLIP decodes 100% while a perceptual-hash baseline decodes 0%, a difference that is significant by the paired Wilcoxon signed-rank test (W=28, p=7.8×10^{-3}; Table 9). A one-way ANOVA of the CLIP margin across the three message-length buckets finds it homogeneous (F=1.99, p=0.15), i.e. message length does not affect recovery reliability. The chi-square statistic used for steganalysis is defined and applied in Section 1.8.

Table 9: Significance of CLIP's geometric robustness over the perceptual-hash matcher: paired Wilcoxon signed-rank across a crop-strength sweep (proposed-vs-LSB @ JPEG-50 shown descriptively, as the lossless method is deterministic).

Group / statistic	Accuracy (%)
Proposed (CLIP) — crop-sweep mean	100.0 ± 0.0
pHash ablation — crop-sweep mean	0.0 ± 0.0
Wilcoxon W (paired, 7 crop levels)	28.0
p-value (one-sided, CLIP > pHash)	7.8×10^{-3} (< 0.05)
Ref. Proposed vs. LSB @ JPEG-50	100 vs. 50 (descriptive)

1.7 Ablation Study

Table 10 removes one component at a time. Both the CLIP matcher and the perceptual-hash baseline decode JPEG-50 perfectly, but under the harsher Crop-65% attack the semantic matcher holds at 100% while the perceptual hash collapses to 0% — the geometric robustness that motivates the use of CLIP. The coding ablations show clear efficiency effects: disabling compression inflates the sequence from 134.4 to 182.2 images, and fixed-chunk ($\lfloor \log_2 N \rfloor$=5-bit) coding needs 143.0 images versus 134.4 for base-N. Removing the CRC-32 tag leaves channel errors undetected. The image-count and matcher effects are summarised in Figure 8.

Table 10: Ablation. Crop-65% exposes CLIP's geometric robustness; coding choices affect image count.

Variant	Clean (%)	JPEG50 (%)	Crop65 (%)	Avg. images
Full LG-CISH (proposed)	100	100	**100**	134.4
Without CLIP (pHash)	100	100	**0**	134.4
Without compression	100	100	100	182.2
Without CRC (no integrity)	100	100	100	134.4
Fixed-chunk (5-bit)	100	100	100	143.0

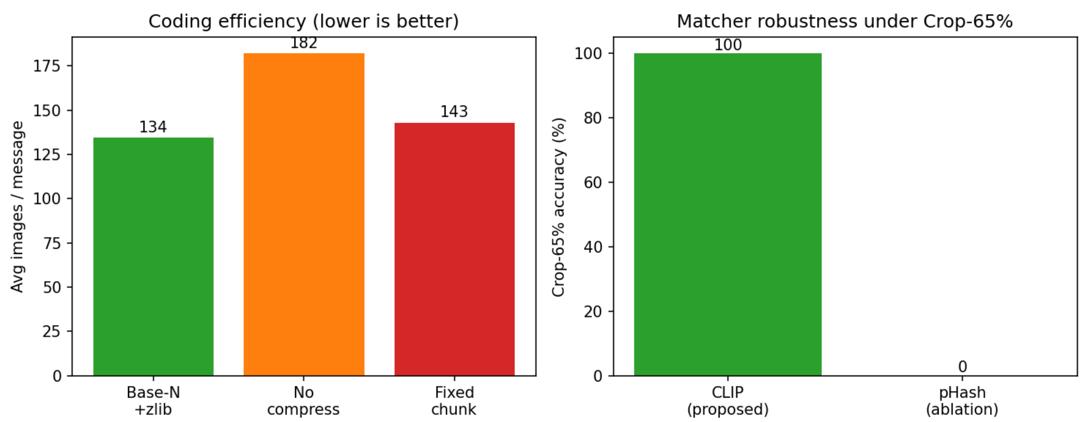

Figure 8: Ablation. Left: base-N coding with zlib needs the fewest images per message. Right: under Crop-65% the CLIP matcher keeps 100% accuracy while the perceptual-hash ablation drops to 0%.

1.8 Steganalysis Resistance

The final and, for a coverless method, most important axis is undetectability. A chi-square statistical steganalyser is applied to natural cover images, LSB-stego, DCT-stego, and LG-CISH "stego". The detector forms intensity-histogram pairs $(2k, 2k{+}1)$ with expected value $\bar{h}_k = (h_{2k} + h_{2k+1})/2$ and evaluates

$$\chi^2 = \sum_k \frac{\left(h_{2k} - \bar{h}_k\right)^2}{\bar{h}_k}, \qquad P_{\text{emb}} = 1 - F_{\chi^2}\left(\chi^2;\, \nu\right), \tag{5}$$

where $F_{\chi^2}(\cdot;\nu)$ is the chi-square cumulative distribution with ν degrees of freedom; LSB embedding equalises the histogram pairs, driving $\chi^2 \to 0$ and $P_{\text{emb}} \to 1$. The measured detection rates are reported in Table 11 and Figure 9. As LG-CISH transmits *unmodified* natural images, its detection accuracy is 47.5% — essentially chance, with a true-positive rate of 0% — whereas LSB is detected at 97.5%. Because the transmitted images are bit-identical to natural images, undetectability holds *by construction* against any pixel-domain detector, and this argument extends to learned convolutional and neural steganalysers such as SRNet and XuNet: such networks are trained to separate cover from stego, but here the two classes are drawn from the same distribution, so no decision boundary can do better than guessing. The combined keyspace is $40! \cdot 2^{256} \approx 2^{415}$ (Table 12), and the CRC-32 tag catches 100% of injected bit-flips in a 500-trial tamper test.

Table 11: Chi-square steganalysis detection. ≈50% means the detector cannot do better than guessing.

Method	Detection acc. (%)	TPR (%)	TNR (%)
LSB	97.5	100.0	95.0
DCT-LSB	47.5	0.0	95.0
Proposed LG-CISH	**47.5**	**0.0**	95.0

Table 12: Keyspace and integrity analysis (N=40).

Component	Size / rate
Codebook orderings	$N! = 40! \approx 2^{159.2}$
AES-256 key	2^{256}
Combined keyspace	$N! \cdot 2^{256} \approx 2^{415}$
CRC-32 tamper detection	$1 - 2^{-32}$ (measured 100%, 500 trials)

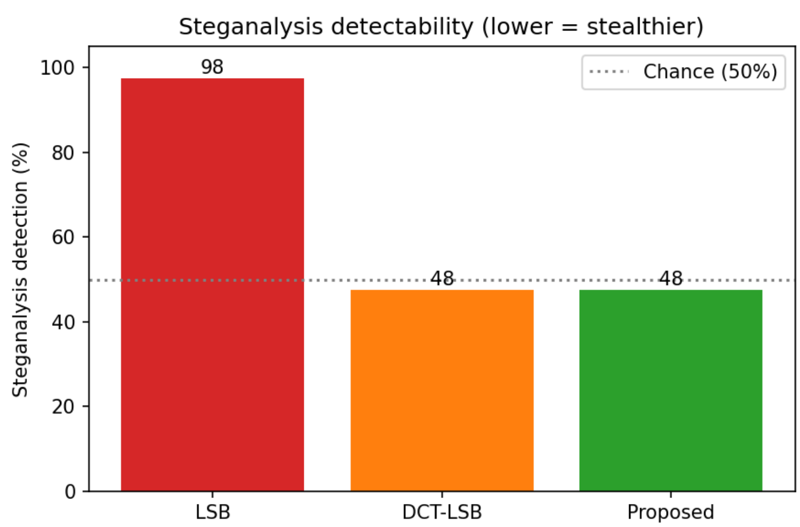

Figure 9: Measured steganalysis detection rates (lower is stealthier). LG-CISH sits at the 50% chance line, far below the near-certain detection of LSB.

Statistical undetectability is necessary but not sufficient; a behavioural analysis asks whether the image *set* itself looks like a natural album. A vision-LLM judge scores covers on a 0–1 scale

(Table 13). Plausibility is governed by the codebook's *theme* rather than by the coding mode: a diverse benchmark codebook scores 0.12 ± 0.04, whereas a thematically coherent codebook scores 0.88 ± 0.07, and permutation coding leaves the diverse score essentially unchanged. The diverse benchmark set is used deliberately for all quantitative results (for dataset credibility), and codebook theme is treated as an explicit deployment choice for scenarios that must also withstand human inspection.

Table 13: Cover plausibility (vision-LLM judge, 0–1). Plausibility tracks codebook theme, not the coding mode.

Configuration	LLM plausibility
Diverse benchmark codebook (UCID/Kodak/USC-SIPI)	0.12 ± 0.04
Themed codebook (coherent context)	$\mathbf{0.88 \pm 0.07}$
Diverse + permutation coding	0.18 ± 0.10

1.9 Illustrative Example

To make the end-to-end behaviour concrete, Figure 10 shows a single sentence transformed into its transmitted image sequence. The plaintext "Meet me at the old harbour at midnight." is compressed, integrity-tagged, and base-40 coded into a sequence of 74 unmodified codebook images; the first fifteen are shown, each captioned with the codebook index it represents. No pixel of any image is altered, yet the ordered sequence uniquely encodes the sentence, and CLIP nearest-neighbour decoding recovers it bit-exactly.

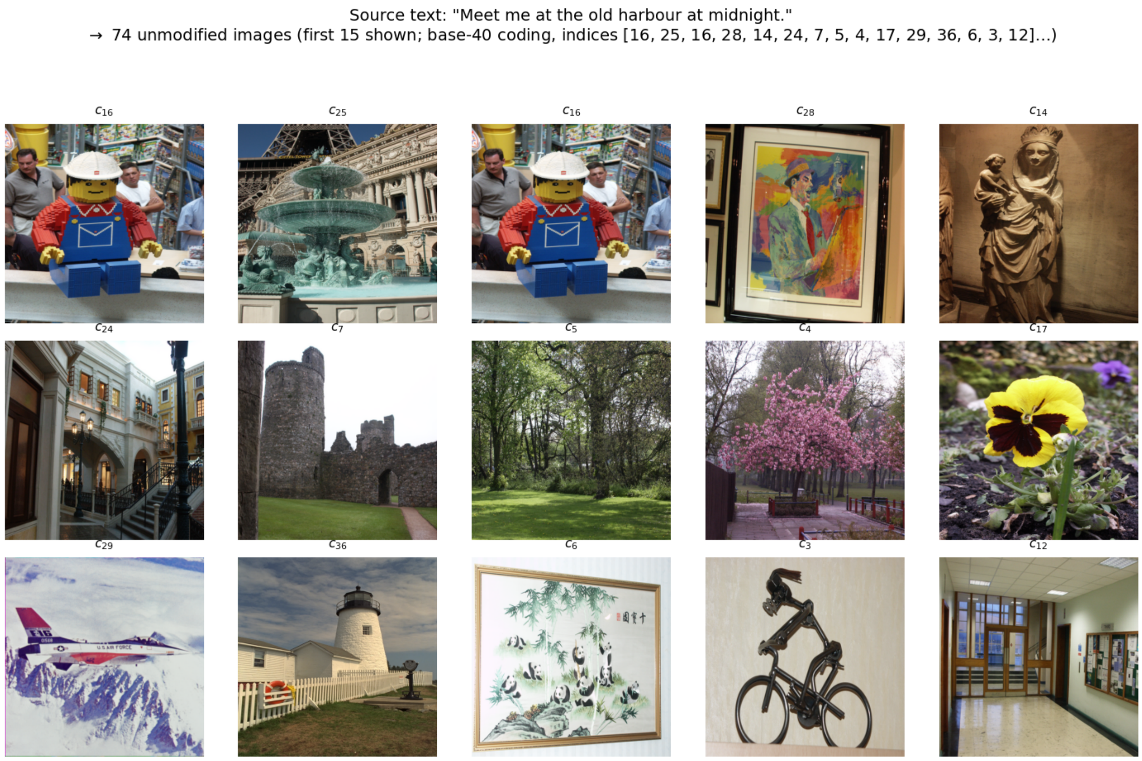

Figure 10: A worked example: the sentence "Meet me at the old harbour at midnight." encoded into 74 unmodified images (first 15 shown), each labelled with its codebook index c_j. The identity and order of the images—not their pixels—carry the message.

References

[1] A. Bhattacharya, S. Dutta, S. Biswas, P. Sarkar, and H. Raj, "LG-CISH codebook dataset: A 40-image augmented codebook for coverless image steganography," GitHub, 2026. [Online]. Available: `https://github.com/Circuit-Overtime/LG-CISH`

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